

Ayush Kashyap — Shipping AI Into a Workflow People Actually Trust

Senior Software Engineer / Tech Lead · Healthtech · ayushkashyaptx@gmail.com · ayushk01.github.io · linkedin.com/in/ayushk01

THE PROBLEM

Doctors lose hours a day to documentation. The obvious fix — record the consultation, transcribe it, have an LLM write the note — is a demo in a week and a failed product in a month. I built the version that survived contact with real clinics.

WHAT I BUILT

- **Ambient AI medical scribe** in production: an AI that drafts structured clinical notes from the consultation for clinician review and sign-off, inside a multi-tenant EHR serving **2,000+ clinics** and hundreds of thousands of patients.
- **Speech pipeline**: speech-to-text for real-world Indian clinic audio (noise, code-switched Hindi-English, overlapping speakers), including speech-model fine-tuning to close the gap generic models left.

THE THREE DECISIONS THAT MADE IT WORK

1. Optimise for “fast to correct”, not “never wrong.” Nothing persists until a clinician signs. That single constraint converts an unsolvable accuracy problem into a solvable editing-speed problem — and it is why the product was safe to ship.

2. Treat fields by blast radius, not uniformly. A hallucinated adjective in prose is noise; a hallucinated dosage or a flipped negation (“no history of diabetes”) is harm. High-risk fields deserve stricter checks than narrative text.

3. Judge it by real use, not by demos. A benchmark score says little about whether a busy clinician keeps using the tool a week later. Adoption in real clinics, not a curated test set, was the bar.

LATENCY IS THE PRODUCT

Adoption did not turn on model quality alone. If transcription lags the conversation, the tool stops being ambient and becomes a distraction. Much of the real engineering was streaming, latency budgets and graceful degradation — the unglamorous layer that decides whether an AI feature gets used twice.

HOW I WORK WITH AI DAY TO DAY

Agentic coding tools are core workflow, not novelty: Claude Code, MCP servers and custom agent harnesses for scaffolding, refactors, migrations and review — with architecture, correctness and the final judgement call staying mine. The same principle as the product: let AI compress idea → shipped, keep a human on anything where being wrong has a cost.

STACK

TypeScript · JavaScript · Python · React · React Native · Node.js / Express · MySQL · Redis · Socket.io · AWS · Docker/K8s · FHIR R4 / HL7